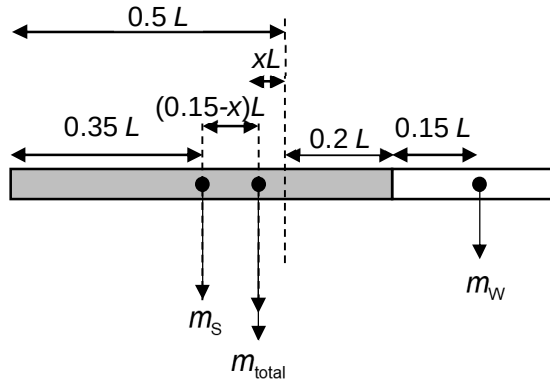


7



Pivot about centre of mass.

By POM,

$$\begin{aligned}\sum \curvearrowright \text{moments} &= (0.35 + x)Lm_w g \\ &= (0.35 + x)Lg(\rho_w A(0.3L))\end{aligned}$$

$$\begin{aligned}\sum \curvearrowright \text{moments} &= (0.15 - x)Lm_s g \\ &= (0.15 - x)Lg(\rho_s A(0.7L)) \\ &= (0.15 - x)Lg(2\rho_w A(0.7L))\end{aligned}$$

$$(0.35 + x)(0.3) = (0.15 - x)(1.4)$$

$$x = 0.062$$

8. upthrust is independent of speed of object and does not act in the direction that opposes motion